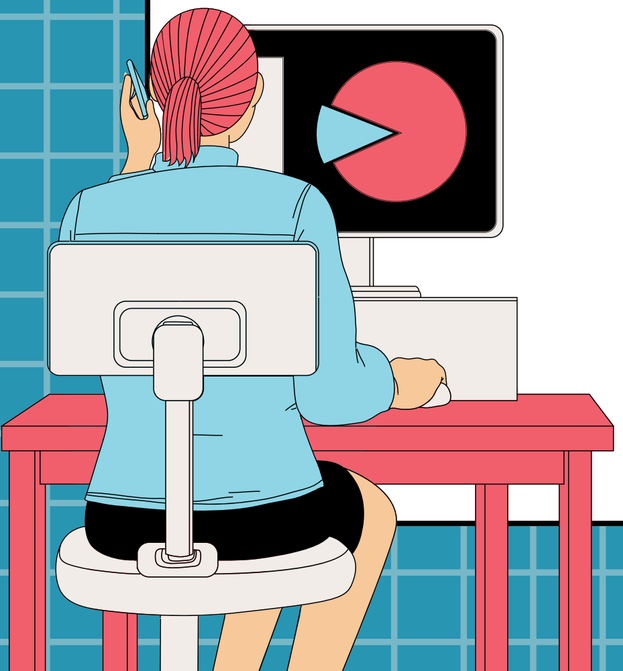
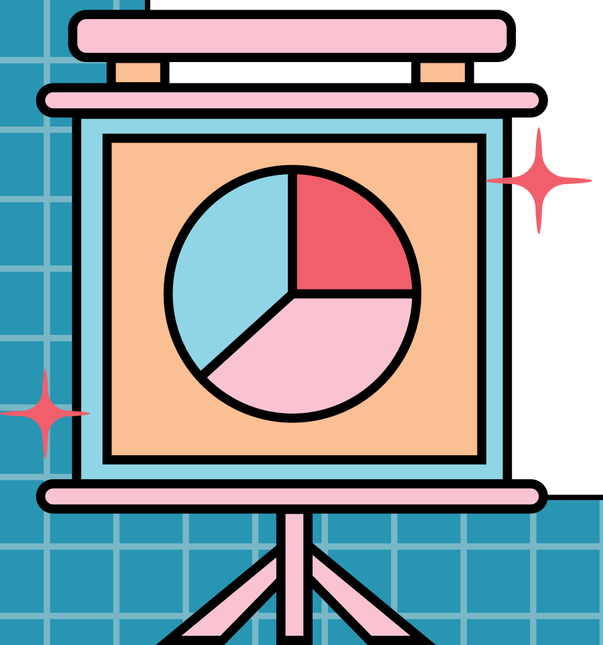
THE IMPLEMENTATION OF BIG DATA PROCESSING USING APACHE PYSPARK FOR WEEKLY SALES FORECASTING AT WALMART



Presented by: Group 6

BUSINESS & BIG DATA JUSTIFICATION



BACKGROUND

Walmart, as one of the largest global retail networks, generates massive and diverse weekly sales data. Accurate forecasting is vital to keep shelves stocked, manage inventory, and satisfy customers, with studies confirming its importance for supply chain decisions like promotions and distribution (Salamena, Fataruba, & Tappa, 2025). However, Walmart struggles to predict demand: overstock wastes resources, while stockouts frustrate customers. Research shows poor forecasting leads to financial losses, highlighting the need for smarter, data-driven approaches (Said et al., 2026; Kabir et al., 2026).



BIG DATA CHARACTERISTIC

Walmart Dataset as Big Data

The Walmart dataset qualifies as big data based on the 5 V's framework:

- Volume – Large quantity of data across stores, departments, and time periods
- Variety – Diverse data types: numerical, categorical, and temporal (weekly sales, temperature, fuel price, CPI, unemployment, holiday status, markdowns)
- Velocity – Sales data updated weekly, requiring fast and continuous processing
- Veracity – Contains missing values in certain columns, necessitating data cleaning for accurate analysis
- Value – Useful for sales forecasting and supporting business decision-making



BUSINESS MOTIVATION

- Accurate sales forecasting is important for Walmart to maintain operational efficiency in a highly dynamic retail environment. By analyzing large-scale sales data, Walmart can optimize inventory management, reduce storage and distribution costs, avoid overstocking or stockouts, and improve customer satisfaction.

- The use of Big Data technologies such as Apache Spark allows faster processing of sales information, enabling more effective planning and quicker responses to changing customer demand

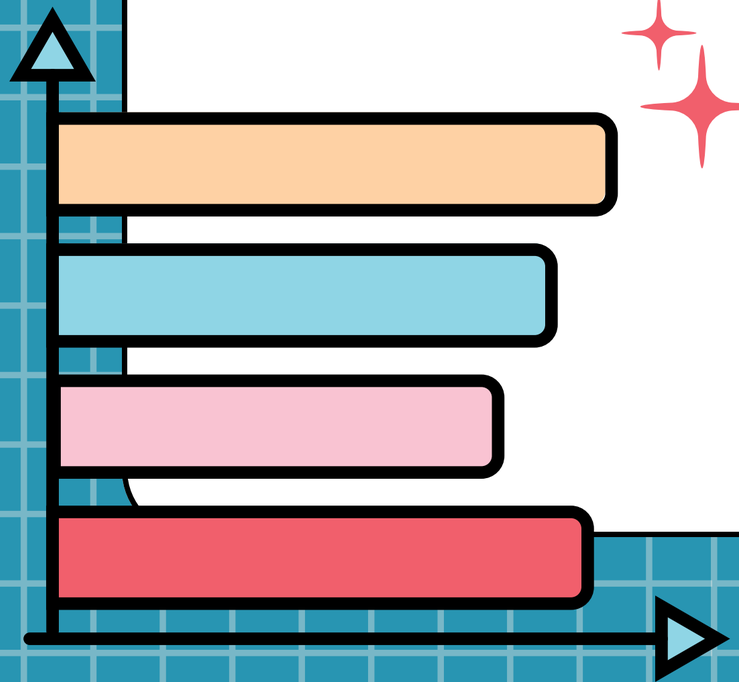


WHY BIG DATA?

Challenge	Traditional Approach	Big Data Solution
Volume	Excel crashes with >1M rows	Spark processes 421,570+ rows seamlessly
Processing Speed	Single-core, sequential	Distributed parallel processing
Complex Analytics	Limited statistical functions	Advanced ML algorithms (Random Forest, etc.)
Scalability	Manual expansion required	Auto-scaling with data growth



BIG DATA ARCHITECTURE DESIGN



BIG DATA OVERVIEW

Data Source

Dataset Walmart Sales Forecasting from Kaggle

Ingestion Layer

Batch Processing

Storage Layer

CSV & Parquet

Processing Layer

Apache Spark & PySpark

Visualization Layer

Matplotlib & Seaborn in Google Colab & PowerBI

Monitoring & security Layer

Access & performance tracking

INGESTION & STORAGE LAYER

INGESTION LAYER

- **Method**
  Batch Ingestion
- **Process**
  Kaggle Dataset → Google Colab → PySpark → Spark DataFrame
- **Technical Justification**
  1. Static historical file
  2. Easy implementation
  3. No real time processing needed
  4. No additional infrastructure needed
- **Reason**
  Suitable for batch processing

STORAGE LAYER

- **Storage Flow**
  Raw Storage → Processed Storage
- **Technical Justification**
  1. Structured data
  2. Medium sized dataset
  3. Low update frequency
- **Technical Justification**
  1. Dataset is not particularly large
  2. Data is not real-time
  3. Structured tabular format
  4. Easy for analysis
- **Architecture**
  HDFS / Data Lake → Data Warehouse

PROCESSING AND VISUALIZATION

PROCESSING / ANALYTICS ENGINE

- The Walmart Sales Forecasting dataset was processed using Apache Spark and PySpark to support large scale data analysis efficiently.
- The preprocessing stage included checking missing values, fixing data formats, transforming date columns, and grouping data based on stores and departments.
- Variables such as weekly sales, fuel price, temperature, CPI, unemployment, and markdown data were analyzed to support the forecasting process.

VISUALIZATION LAYER

- The processed data was presented using graphs and charts to make the analysis results easier to understand.
- Weekly sales trends and patterns across different stores and departments were visualized clearly.
- Additional factors such as holidays, fuel prices, CPI, unemployment, and markdown promotions were included to analyze their influence on sales performance.

MONITORING & SECURITY LAYER

MONITORING LAYER

To monitor anomalies, performance stability and system health

- **Monitoring Components**

1. Infrastructure (CPU, Memory, Storage): Monitor server workload in Google Colab Resource Monitor
2. Data Pipeline Health & Failures: Monitor the execution process of each code cell in Google Colab.
3. Model Performance (WMAE Drift): Perform periodic evaluations of the RMSE and R^{2} values of PySpark code results.
4. API Latency & Dashboard Uptime: Checking refresh rates and dataset connections on the Power BI Service.

SECURITY LAYER

To ensure that the system is prone from cyber attacks and prevent unauthorized access

- **Security components**

1. Data protection: Using the Parquet file format for data storage.
2. Access control: Configuring Share Permissions in Google Drive/Colab and Power BI.
3. Audit logs: Utilizing the Version History feature in Google Colab.
4. Network security: Relying on Google Cloud's standard encryption since the work is conducted within the Google ecosystem.

INFRASTRUCTURE ANALYSIS DESIGN

- Walmart Sales Forecasting dataset is structured, has low latency, and moderate volume, a cloud-based data warehouse is the best choice.
- For the deployment model, this project used a public cloud via Google Colab. This choice was made because it was more cost-effective, easily accessible, and enabled collaboration with other teams.



IMPLEMENTATION



DATASET OVERVIEW



```
df_train head:
   Store  Dept        Date  Weekly_Sales  IsHoliday
0      1     1  2010-02-05      24924.50      False
1      1     1  2010-02-12      46039.49       True
2      1     1  2010-02-19      41595.55      False
3      1     1  2010-02-26      19403.54      False
4      1     1  2010-03-05      21827.90      False
Jumlah baris train: 421570

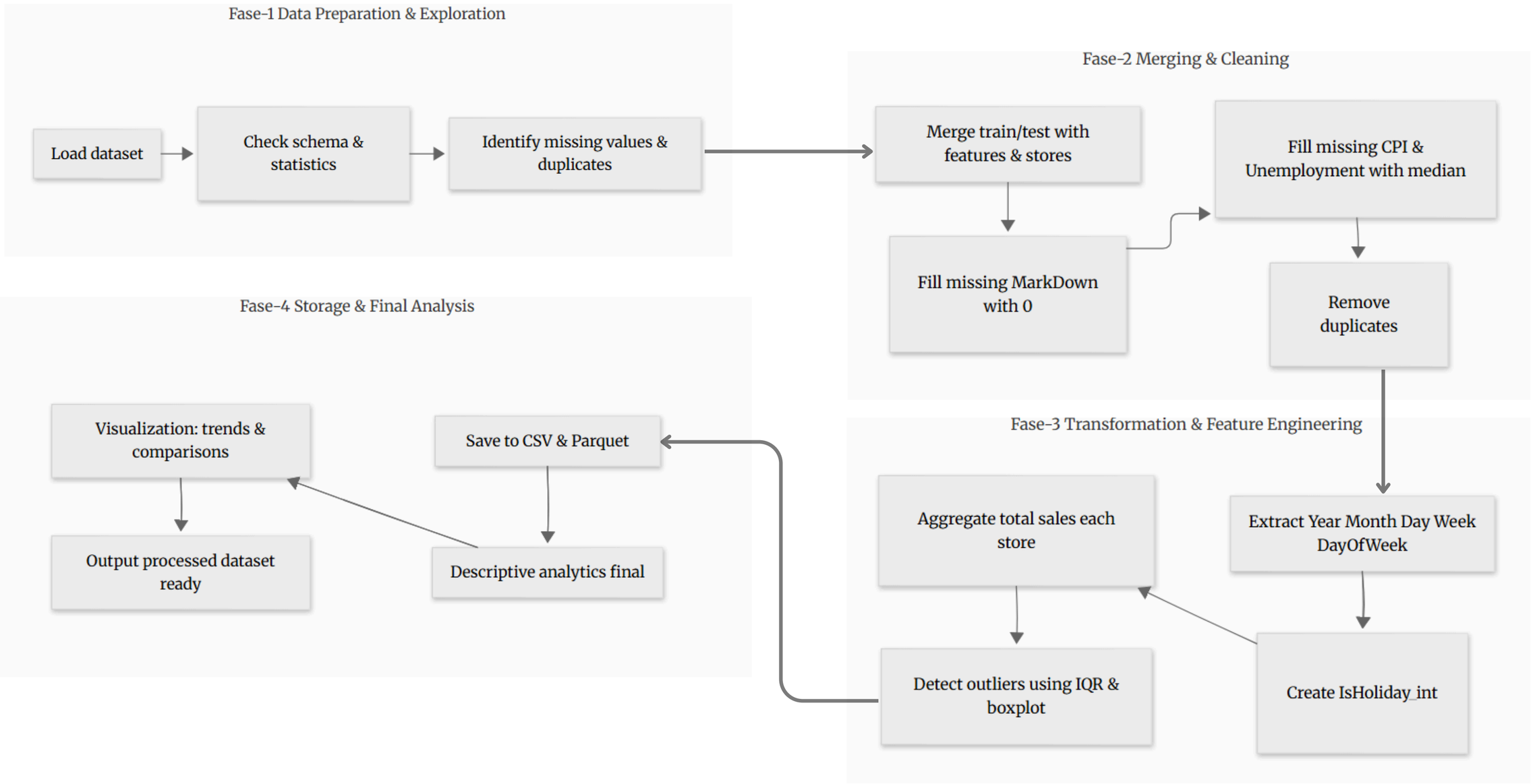
df_test head:
   Store  Dept        Date  IsHoliday
0      1     1  2012-11-02      False
1      1     1  2012-11-09      False
2      1     1  2012-11-16      False
3      1     1  2012-11-23       True
4      1     1  2012-11-30      False
Jumlah baris test:  115064

df_features head:
   Store        Date  Temperature  Fuel_Price  MarkDown1  MarkDown2  \
0      1  2010-02-05        42.31       2.572        NaN        NaN
1      1  2010-02-12        38.51       2.548        NaN        NaN
2      1  2010-02-19        39.93       2.514        NaN        NaN
3      1  2010-02-26        46.63       2.561        NaN        NaN
4      1  2010-03-05        46.50       2.625        NaN        NaN

   MarkDown3  MarkDown4  MarkDown5         CPI  Unemployment  IsHoliday
0        NaN        NaN        NaN  211.096358         8.106      False
1        NaN        NaN        NaN  211.242170         8.106       True
2        NaN        NaN        NaN  211.289143         8.106      False
3        NaN        NaN        NaN  211.319643         8.106      False
4        NaN        NaN        NaN  211.350143         8.106      False
Jumlah baris features: 8190

df_stores head:
   Store Type    Size
0      1    A  151315
1      2    A  202307
2      3    B   37392
3      4    A  205863
4      5    B   34875
Jumlah baris stores: 45
```

DATA PROCESSING



DATA CLEANING & FUTURE ENGINEERING

DATA CLEANING

- Merged train, features, and stores datasets using PySpark
- Missing MarkDown values were filled with 0
- CPI and Unemployment handled using median imputation
- Duplicate checking performed before analytics process

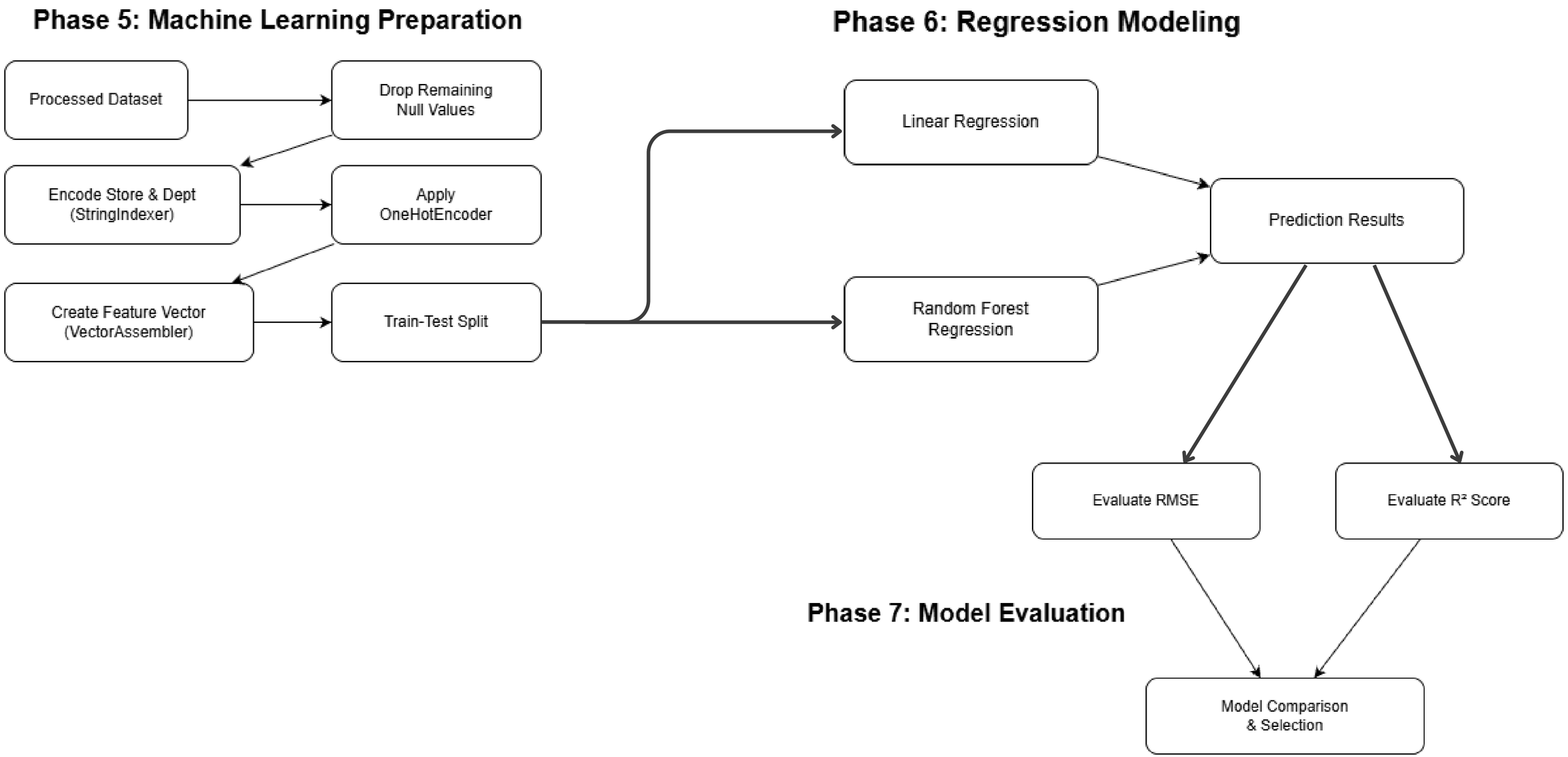
FEATURE ENGINEERING

- Extracted Year, Month, Week, and DayOfWeek from Date
- Converted IsHoliday into numerical format
- Generated features for forecasting and machine learning

OUTLIER DETECTION

- Used IQR and boxplot analysis
- Checked abnormal values in numerical features before modeling

BIG DATA ANALYTICS



ML PREPARATION & REGRESSION MODELING

MACHINE LEARNING PREPARATION

- Applied StringIndexer for categorical indexing
- Used OneHotEncoder to transform categorical variables
- Combined features using VectorAssembler
- Split dataset into training and testing data

REGRESSION MODELS

- Linear Regression
- Random Forest Regression

EVALUATION METRICS

- RMSE (Root Mean Square Error)
- R^2 Score

```
# 2. Encode Store
store_indexer_model = StringIndexer
                      (inputCol="Store", outputCol="Store_index").fit(df_train_clean)
df_train_clean = store_indexer_model.transform(df_train_clean)
```

```
# 4. One Hot Encoding
encoder_model = OneHotEncoder(inputCols=["Store_index", "Dept_index"],
    outputCols=["Store_vec", "Dept_vec"]).fit(df_train_clean)
df_train_clean = encoder_model.transform(df_train_clean)
```

```
# 6. Vector assembler
assembler = VectorAssembler(inputCols=feature_cols_final,
                            outputCol='features')
data_train = assembler.transform(df_train_clean)
```

MODEL EVALUATION RESULTS

Model	R^2	RMSE
Linear Regression	94.22%	12.33%
Random Forest Regression	81.70%	21.94%

EVALUATION RESULTS

- Linear Regression produced lower RMSE compared to Linear Regression
- Higher R^2 score indicates better prediction capability
- Model successfully captured sales patterns from historical data

FINAL MODEL SELECTION

- Linear Regression chosen as the final forecasting model
- More effective for handling complex retail sales behavior
- Suitable for Walmart weekly sales prediction per stores

CHALLENGES & LIMITATIONS

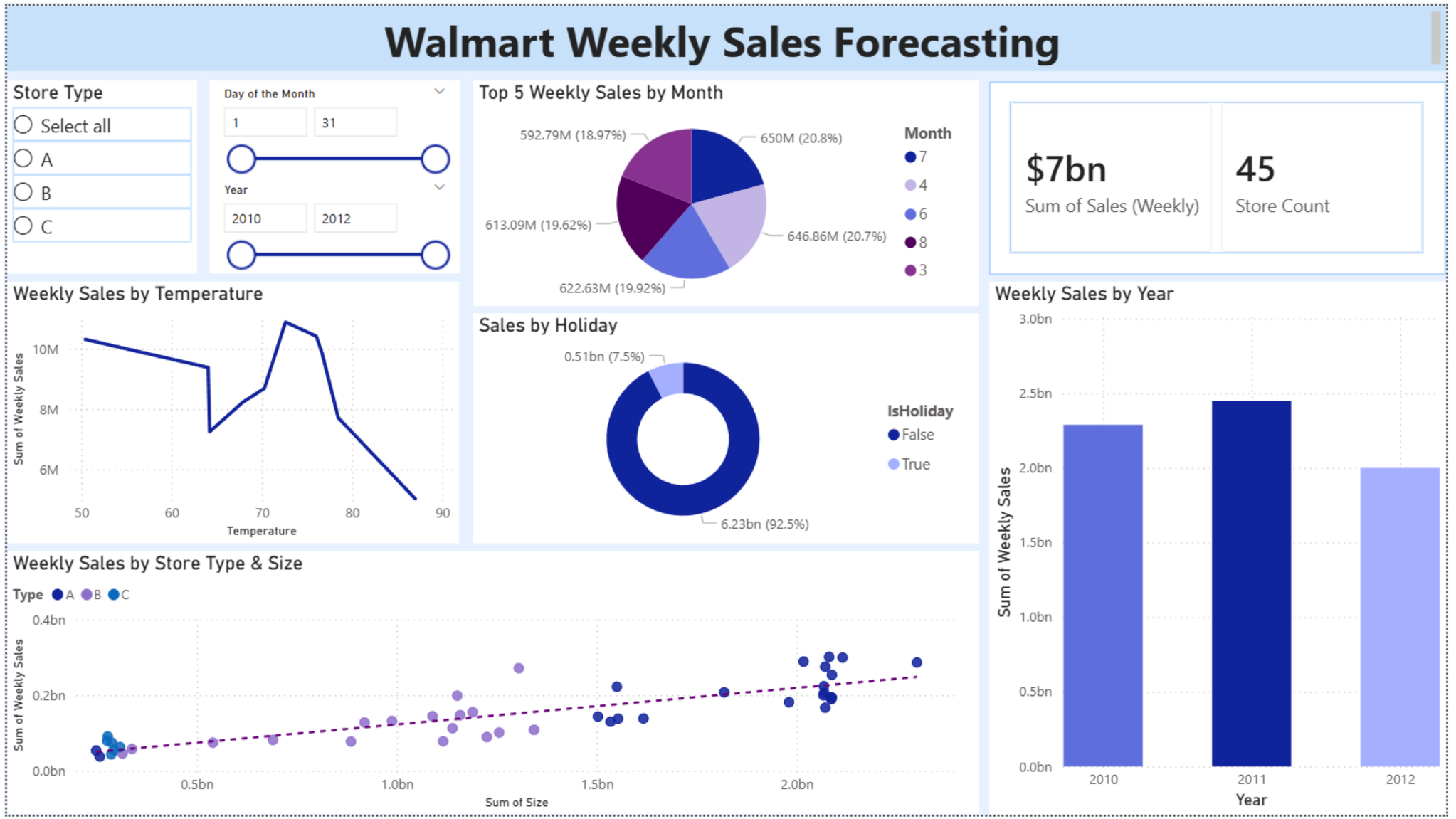
CHALLENGES

- Several promotional columns contained many null values and required imputation
- Free-tier environment restricted memory and processing capability
- Data merging, transformation, and model training required longer execution time
- Some prediction results were still far from actual sales values

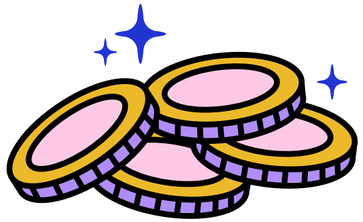
LIMITATIONS

- External factors such as weather trends and economic conditions were not fully explored
- Time-based split may affect model generalization
- Model performance can still be improved using advanced algorithms

POWER BI

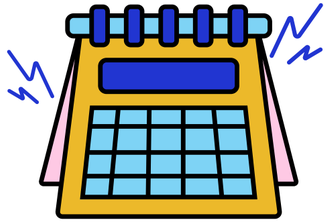

Walmart Weekly Sales Forecasting
Store Type
Select all
A
B
C
Day of the Month
1
31
Year
2010
2012
Top 5 Weekly Sales by Month
592.79M (18.97%)
650M (20.8%)
613.09M (19.62%)
646.86M (20.7%)
622.63M (19.92%)
Month
7
4
6
8
3
$7bn
Sum of Sales (Weekly)
45
Store Count
Weekly Sales by Temperature
Sum of Weekly Sales
10M
8M
6M
50
60
70
80
90
Temperature
Sales by Holiday
0.51bn (7.5%)
6.23bn (92.5%)
IsHoliday
False
True
Weekly Sales by Year
3.0bn
2.5bn
2.0bn
1.5bn
1.0bn
0.5bn
0.0bn
Sum of Weekly Sales
2010
2011
2012
Year
Weekly Sales by Store Type & Size
Type A B C
0.4bn
0.2bn
0.0bn
Sum of Weekly Sales
0.5bn
1.0bn
1.5bn
2.0bn
Sum of Size

BUSINESS INSIGHT



Revenue Dominance

$7 billion of total sales, most of the revenue is heavily driven by type A stores.



Golden Season

Months March - April & June - August have the biggest sales. Sales peak between 70˚F - 75˚F.



Daily Operation

92.5% of total sales come from non-holiday weeks, 7.5% from holiday weeks.

Power BI:

https://app.powerbi.com/links/X6faEcIV3T?ctid=3485b963-82ba-4a6f-810f-b5cc226ff898&pbi_source=linkShare

FUTURE IMPROVEMENT

- Use better computing infrastructure like native Spark clusters or Google Colab Pro for deeper exploration.
- The use of more accurate validation techniques can provide more accurate estimates of model performance.
- Adding data from external sources, such as regional economic conditions, national holidays, and weather data for each region, will further improve the predictive model's accuracy due to the data's greater support.

CONCLUSION

- PySpark enabled scalable data cleaning, transformation, and analytics
- Random Forest achieved the best forecasting performance
- Power BI visualization supported business insight generation

THANK YOU

